

Q1.

The time in hours to failure of a component may be modelled by an exponential distribution with parameter $\lambda = 0.025$
In a manufacturing process, the machine involved uses one of these components continuously until it fails.

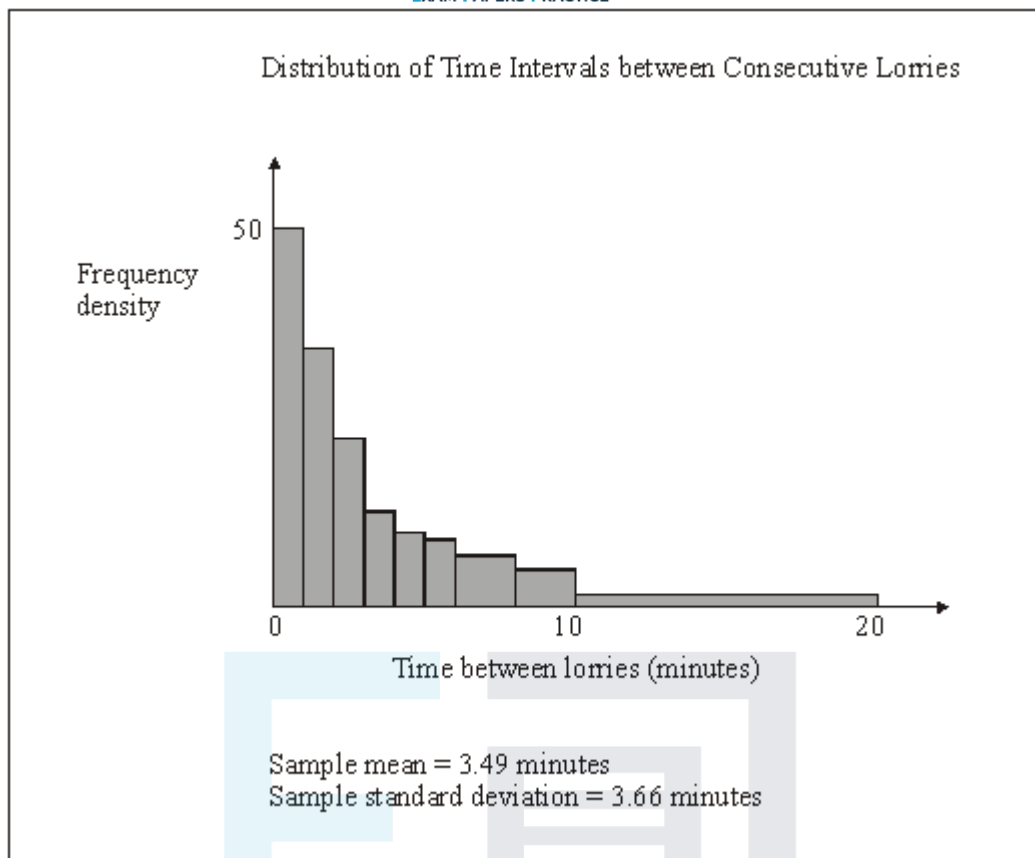
The component is then immediately replaced.

- (a) Write down the mean time to failure for a component. (1)
 - (b) Find the probability that a component will fail during a 12-hour shift. (1)
 - (c) A component has not failed for 30 hours. Find the probability that this component lasts for at least another 30 hours. (2)
 - (d) Find the probability that a component does **not** fail during 4 consecutive 12-hour shifts. (3)
 - (e) (i) State the distribution that can be used to model the number of components that fail during one hour of the manufacturing process. (2)
 - (ii) Hence, or otherwise, find the probability that no components fail during 5 consecutive 12-hour shifts. (2)
- (Total 11 marks)**

Q2.

The parents of children at a village primary school are concerned about the frequency of lorries passing the school. The Parents' Association organises a survey. Over a three-week period, volunteers stand outside the school, at randomly selected times during the school day, and record the time interval between consecutive lorries driving past in either direction.

When the survey is complete, the secretary of The Parents' Association summarises the data by drawing a histogram and calculating the mean and standard deviation of the time intervals between consecutive lorries. The results are shown below.



- (a) Using the above summary of data, state reasons why it might be appropriate to use an exponential distribution to model the time intervals between consecutive lorries. (2)
- (b) The distribution of the time interval, T minutes, between consecutive lorries is to be modelled as an exponential distribution with distribution function:
- $$P(T \leq t) = F(t) = \begin{cases} 1 - e^{-0.3t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$
- (i) Find the mean value of T . (1)
- (ii) Find the probability that the time interval between consecutive lorries is less than one minute. (2)
- (iii) A school crossing patrol lady helps children to cross the road outside the school. One morning she steps out into the road to stop the traffic exactly one minute after a lorry has passed. She then holds up the traffic for exactly 45 seconds. Find the probability that no lorries are waiting when she allows the traffic to move. (3)
- (iv) Calculate the median time interval between consecutive lorries. (3)
- (Total 11 marks)**

Q3.

The times, T minutes, between successive telephone calls to a small business can be assumed to be exponentially distributed with mean 10.

- (a) (i) Write down, in full, the probability density function of T . (2)
- (ii) Using integration, obtain an expression, valid for $t \geq 0$, for the distribution function of T . (3)
- (b) A call is received at 11.00 am.
- Find the probability that the next call is received:
- (i) before 11.15 am; (2)
- (ii) between 11.10 am and 11.20 am. (2)

(Total 9 marks)

Q4.

John travels to work by bus and he arrives at the bus stop at exactly 8 am every morning. From past experience, he knows that the next bus arrives at X minutes past 8, where X is an exponentially distributed random variable with mean 10.

- (a) Write down the probability density function of X . Deduce that the distribution function of X is given, for $x \geq 0$, by

$$F(x) = 1 - e^{-\frac{x}{10}}$$

(4)

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- (b) Calculate the probability that no bus has arrived by 8.15 am. (2)
- (c) Given that no bus has arrived by 8.15 am, calculate the probability that the next bus arrives before 8.20 am. (3)

(Total 9 marks)

Q5.

The lifetime, T hours, of a certain electrical component is exponentially distributed with mean 800.

- (a) (i) State, in full, the probability density function of T . (2)
- (ii) Write down the numerical value of the standard deviation of T . (1)

- (iii) Calculate the probability that one of these components, chosen at random, is still working after 1000 hours. (3)
- (iv) Given that one of these components is still working after 1000 hours, calculate the probability that it is still working after a further 1000 hours. (3)
- (b) A piece of equipment contains four of these components. The design is such that it is still functional when at least two of the four components are working. Calculate the probability that the piece of equipment is still functional after 1000 hours. (4)
- (Total 13 marks)

Q6.

Trains stop at a particular station for a minimum of 1 minute.

The time, T minutes, **in excess of 1 minute**, that trains stop at the station may be modelled by an exponential distribution with mean 2.

- (a) State the probability that a train stops at the station for at least 1 minute. (1)
- (b) Calculate the probability that a train stops at the station for more than 5 minutes. (3)
- (c) A train has been stopped at the station for 3 minutes.
Calculate the probability that it stops at the station for **a total of** less than 5 minutes. (4)
- (d) A random sample of five trains each stop at the station for more than 5 minutes.
Indicate why this casts doubt on the above model. (2)

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(Total 10 marks)